Chapter 4: Elasticity and Its Application

---

1. PRICE ELASTICITY OF DEMAND (PED)

Definition:

Price Elasticity of Demand measures the responsiveness of quantity demanded to a change in price

Basic Formula:

$$\boxed{PED = \frac{\%\text{ Change in } Q_d}{\%\text{ Change in } P}}$$

OR

$$\boxed{PED = \left|\frac{\Delta Q_d / Q_d}{\Delta P / P}\right|}$$

Variable Definitions: | Symbol | Full Form | Unit | Description | |——|———|——|————-| | PED | Price Elasticity of Demand | unitless | Responsiveness measure | | ΔQd | Change in Quantity Demanded | units | Qd - Qd | | Qd | Quantity Demanded | units | Original quantity demanded | | ΔP | Change in Price | Rs., $ | P - P | | P | Price | Rs./unit, $/unit | Original price | | | | | Absolute value | - | Always report PED as positive |

When to Use:

Measuring how sensitive demand is to price changes
Pricing strategy decisions
Predicting revenue impacts
MOST IMPORTANT ELASTICITY - appears in almost every exam

---

2. MIDPOINT METHOD (PREFERRED)

Formula:

$$\boxed{PED = \frac{\left|\frac{Q_2 - Q_1}{(Q_2 + Q_1)/2}\right|}{\left|\frac{P_2 - P_1}{(P_2 + P_1)/2}\right|}}$$

Simplified:

$$\boxed{PED = \frac{|Q_2 - Q_1|}{|P_2 - P_1|} \times \frac{(P_2 + P_1)/2}{(Q_2 + Q_1)/2}}$$

Variable Definitions: | Symbol | Full Form | Description | |——|———|————-| | Q | Initial Quantity | Quantity at first point | | Q | Final Quantity | Quantity at second point | | P | Initial Price | Price at first point | | P | Final Price | Price at second point | | (Q +Q)/2 | Midpoint of Quantity | Average quantity | | (P +P)/2 | Midpoint of Price | Average price |

Why Use Midpoint Method?

Direction-independent: Same answer whether going from A→B or B→A
More accurate: Uses average as base
Exam standard: Most commonly required method

Example 1: Basic PED Calculation

Problem Statement: *"Price rises by 10%. Quantity demanded falls by 15%. Calculate price elasticity of demand."*

Given: - % Change in P = 10% - % Change in Qd = 15%

Solution:

$$PED = \frac{15\%}{10\%} = \boxed{1.5}$$

Interpretation: - PED = 1.5 > 1 → **Elastic demand** - 1% price increase causes 1.5% quantity decrease - Demand is price-sensitive

---

Example 2: Midpoint Method Application

Problem Statement: *"A company sells websites. At P = $200, demand is 12 units. At P = $250, demand is 8 units. Calculate PED using midpoint method."*

Given: | Point | Price (P) | Quantity (Q) | |——-|———–|————–| | A | $200 | 12 | | B | $250 | 8 |

Solution:

Step 1: Calculate midpoints

$$\text{Price Midpoint} = \frac{200 + 250}{2} = \frac{450}{2} = \$225$$

$$\text{Quantity Midpoint} = \frac{12 + 8}{2} = \frac{20}{2} = 10 \text{ units}$$

Step 2: Calculate % changes

$$\%\Delta P = \frac{250 - 200}{225} \times 100 = \frac{50}{225} \times 100 = 22.2\%$$

$$\%\Delta Q = \frac{8 - 12}{10} \times 100 = \frac{-4}{10} \times 100 = -40.0\%$$

(Take absolute value: 40.0%)

Step 3: Calculate PED

$$PED = \frac{40.0\%}{22.2\%} = \boxed{1.8}$$

Interpretation: - PED = 1.8 > 1 → **Elastic demand** - Demand is quite responsive to price changes

---

Example 3: Hotel Rooms Elasticity (Active Learning)

Problem Statement: *"Hotel room demand data: When P = \$70, Qd = 5000 rooms. When P = \$90, Qd = 3000 rooms. Calculate price elasticity of demand."*

Given: - P = \$70, Qd = 5000 - P = \$90, Qd = 3000

Solution:

Midpoints:

$$P_{mid} = \frac{70 + 90}{2} = \$80$$
$$Q_{mid} = \frac{5000 + 3000}{2} = 4000$$

% Changes:

$$\%\Delta P = \frac{90 - 70}{80} \times 100 = \frac{20}{80} \times 100 = 25\%$$

$$\%\Delta Q = \frac{3000 - 5000}{4000} \times 100 = \frac{-2000}{4000} \times 100 = -50\%$$

(Absolute value: 50%)

PED:

$$PED = \frac{50\%}{25\%} = \boxed{2.0}$$

Interpretation: - PED = 2.0 > 1 → **Elastic demand** - Hotel demand is very price-sensitive - 1% price increase → 2% quantity decrease

---

3. CLASSIFICATIONS OF DEMAND ELASTICITY

Summary Table:

Type	PED Value	Interpretation	Demand Curve	Example
Perfectly Inelastic	0	No response to price	Vertical	Life-saving drugs
Inelastic	< 1	Small response	Steep	Necessities (salt, insulin)
Unit Elastic	$= 1$	Proportional response	Medium slope	-
Elastic	> 1	Large response	Flat	Luxuries (cruises)
Perfectly Elastic	∞	Infinite response	Horizontal	Perfect competition

---

1. Perfectly Inelastic (PED = 0)

$$\boxed{PED = 0 \quad \text{(Vertical demand curve)}}$$

Characteristics: - Quantity **unchanged** when price changes - **No substitutes** available - **Absolute necessity**

Example: Insulin for diabetics - Price rises 10% → Quantity demanded changes 0%

---

2. Inelastic Demand (PED < 1)

$$\boxed{0 < PED < 1 \quad \text{(Steep demand curve)}}$$

Characteristics: - % ΔQd < % ΔP - **Few substitutes** - **Necessities**

Examples: Salt, gasoline (short-run), insulin

Key Point: Price ↑ → TR ↑ (Total Revenue increases)

---

3. Unit Elastic (PED = 1)

$$\boxed{PED = 1 \quad \text{(Intermediate slope)}}$$

Characteristics: - % ΔQd = % ΔP - **Proportional response**

Key Point: Price changes → TR **unchanged**

---

4. Elastic Demand (PED > 1)

$$\boxed{PED > 1 \quad \text{(Flat demand curve)}}$$

Characteristics: - % ΔQd > % ΔP - **Many substitutes** - **Luxuries**

Examples: Caribbean cruises, restaurant meals, designer clothes

Key Point: Price ↑ → TR ↓ (Total Revenue decreases)

---

5. Perfectly Elastic (PED = ∞)

$$\boxed{PED = \infty \quad \text{(Horizontal demand curve)}}$$

Characteristics: - Any price increase → Quantity drops to **zero** - Perfect competition

---

4. TOTAL REVENUE AND ELASTICITY

Total Revenue Formula:

$$\boxed{TR = P \times Q}$$

Where: | Symbol | Full Form | Unit | Description | |———|————|——|————-| | TR | Total Revenue | Rs., $ | Total sales income | | P | Price | Rs./unit, $/unit | Selling price per unit | | Q | Quantity | units | Quantity sold |

TR-Elasticity Relationships:

$$\boxed{\begin{cases} \text{If PED} < 1 \text{ (Inelastic):} & P \uparrow \Rightarrow TR \uparrow \\ & P \downarrow \Rightarrow TR \downarrow \\ \\ \text{If PED} > 1 \text{ (Elastic):} & P \uparrow \Rightarrow TR \downarrow \\ & P \downarrow \Rightarrow TR \uparrow \\ \\ \text{If PED} = 1 \text{ (Unit Elastic):} & P \text{ changes} \Rightarrow TR \text{ unchanged} \end{cases}}$$

When to Use:

Predicting revenue impact of price changes
Pricing strategy - should we raise or lower price?
Maximizing revenue

Example 4: Inelastic Demand and TR

Problem Statement: *"Demand is inelastic. Price increases from $1 to $3. Quantity falls from 100 to 80 units. Analyze total revenue impact."*

Given: - P = $1, Q = 100 - P = $3, Q = 80

Solution:

Initial TR:

$$TR_1 = P_1 \times Q_1 = 1 \times 100 = \$100$$

New TR:

$$TR_2 = P_2 \times Q_2 = 3 \times 80 = \$240$$

Change in TR:

$$\Delta TR = 240 - 100 = +\$140 \text{ (Increase)}$$

Conclusion: - Price ↑ → TR ↑ - Confirms **inelastic demand** (PED < 1) - **Strategy:** Raising price increased revenue

---

Example 5: Elastic Demand and TR

Problem Statement: *"Demand is elastic. Price increases from $4 to $5. Quantity falls from 50 to 20 units. Analyze total revenue impact."*

Given: - P = $4, Q = 50 - P = $5, Q = 20

Solution:

Initial TR:

$$TR_1 = 4 \times 50 = \$200$$

New TR:

$$TR_2 = 5 \times 20 = \$100$$

Change in TR:

$$\Delta TR = 100 - 200 = -\$100 \text{ (Decrease)}$$

Conclusion: - Price ↑ → TR ↓ - Confirms **elastic demand** (PED > 1) - **Strategy:** Raising price decreased revenue - **Better:** Lower price to increase revenue

---

5. INCOME ELASTICITY OF DEMAND (YED)

Formula:

$$\boxed{YED = \frac{\text{\% Change in } Q_d}{\text{\% Change in Income}}}$$

$$\boxed{YED = \frac{\Delta Q_d/Q_d}{\Delta Y/Y}}$$

Variable Definitions: | Symbol | Full Form | Unit | Description | |——–|————–|——|—————-| | YED | Income Elasticity of Demand | unitless | Income responsiveness | | ΔQd | Change in Quantity Demanded | units | Qd - Qd | | Qd | Quantity Demanded | units | Original quantity | | ΔY | Change in Income | Rs., $ | Y - Y | | Y | Income | Rs., $ | Original income |

When to Use:

Analyzing how income changes affect demand
Classifying goods (luxury vs necessity vs inferior)
Forecasting demand based on economic growth

YED Classification:

YED Value	Type of Good	Income ↑ Effect	Examples
YED > 1	Luxury (Income Elastic)	Demand ↑↑ (large increase)	Sports cars, furs, fine dining
0 < YED < 1	Necessity (Income Inelastic)	Demand ↑ (small increase)	Food, fuel, clothing, utilities
YED < 0	Inferior Good	Demand ↓	Low-quality goods, public transport
YED = 0	Perfectly Income Inelastic	Demand unchanged	Salt

Example 6: Income Elasticity Calculation

Problem Statement: *"When income is Rs. 500, quantity demanded is 20 units. When income rises to Rs. 600, quantity demanded becomes 25 units. Calculate income elasticity."*

Given: | Income (Y) | Qd | |———–|—–| | Rs. 500 | 20 units | | Rs. 600 | 25 units |

Solution:

$$\Delta Y = 600 - 500 = 100$$
$$\Delta Q_d = 25 - 20 = 5$$

$$YED = \frac{\Delta Q_d/Q_d}{\Delta Y/Y} = \frac{5/20}{100/500}$$

$$YED = \frac{0.25}{0.20} = \boxed{1.25}$$

Interpretation: - YED = 1.25 > 1 → **Luxury good (Income Elastic)** - 1% income increase → 1.25% demand increase - As people get richer, they buy proportionally more of this good

---

6. CROSS-PRICE ELASTICITY OF DEMAND (XED)

Formula:

$$\boxed{XED = \frac{\text{\% Change in } Q_d \text{ of Good A}}{\text{\% Change in Price of Good B}}}$$

$$\boxed{XED = \frac{\Delta Q_{d,A}/Q_{d,A}}{\Delta P_B/P_B}}$$

Variable Definitions: | Symbol | Full Form | Description | |——–|———–|————-| | XED | Cross-Price Elasticity | Relationship between two goods | | ΔQd,A | Change in Qd of Good A | How much demand for A changes | | Qd,A | Quantity Demanded of A | Original demand for good A | | ΔPB | Change in Price of Good B | How much price of B changes | | PB | Price of Good B | Original price of good B |

When to Use:

Determining if goods are substitutes or complements
Analyzing competitive products
Market strategy (pricing related products)

XED Classification:

XED Value	Relationship	Meaning	Examples
XED > 0	**Substitutes**	PB ↑ → Qd,A ↑	Beef & Chicken, Tea & Coffee
XED < 0	**Complements**	PB ↑ → Qd,A ↓	Cars & Gas, Printers & Ink
XED = 0	**Unrelated**	No relationship	Cars & Bread

Example 7: Cross Elasticity (Wheat & Rice)

Problem Statement: *"Wheat is priced at Rs. 1600 for 40 kg, demand is 40 kg. Rice price increases from Rs. 2000 to Rs. 2500. Wheat demand increases to 50 kg. Find cross-price elasticity."*

Given:

Commodity	Price Change	Qd Change
Wheat A	PA = Rs. 1600 (constant)	40 kg → 50 kg
Rice B	Rs. 2000 → Rs. 2500	-

Solution:

$$\Delta Q_{d,A} = 50 - 40 = 10 \text{ kg}$$
$$Q_{d,A} = 40 \text{ kg}$$

$$\Delta P_B = 2500 - 2000 = 500$$
$$P_B = 2000$$

$$XED = \frac{\Delta Q_{d,A}/Q_{d,A}}{\Delta P_B/P_B} = \frac{10/40}{500/2000}$$

$$XED = \frac{0.25}{0.25} = \boxed{1.0}$$

Interpretation: - XED = 1.0 > 0 → Wheat and Rice are **SUBSTITUTES** - Unit elastic cross-price relationship - When rice price ↑ 1%, wheat demand ↑ 1%

---

7. PRICE ELASTICITY OF SUPPLY (PES)

Formula:

$$\boxed{PES = \frac{\% \text{ Change in } Q_s}{\% \text{ Change in } P}}$$

$$\boxed{PES = \frac{\Delta Q_s/Q_s}{\Delta P/P}}$$

Variable Definitions: | Symbol | Full Form | Unit | Description | |——–|————|——|————-| | PES | Price Elasticity of Supply | unitless | Supply responsiveness | | ΔQs | Change in Quantity Supplied | units | Qs - Qs | | Qs | Quantity Supplied | units | Original quantity supplied | | ΔP | Change in Price | Rs., $ | P - P | | P | Price | Rs./unit | Original price |

When to Use:

Measuring how quickly suppliers can respond to price changes
Analyzing production flexibility
Time period analysis (short-run vs long-run)

PES Classification:

PES Value	Type	Interpretation	Context
0	Perfectly Inelastic	Fixed supply	Unique items (Mona Lisa)
< 1	Inelastic	Supply difficult to change	Short-run, agricultural products
= 1	Unit Elastic	Proportional response	-
> 1	Elastic	Supply easy to change	Long-run, manufactured goods
∞	Perfectly Elastic	Infinite supply at price	Perfect competition

---

Example 8A: Supply Elasticity Calculation

Problem Statement: *"When price is Rs. 10, quantity supplied is 20 units. When price rises to Rs. 12, quantity supplied is 24 units. Calculate PES."*

Given: | Price (P) | Qs | |———–|—–| | Rs. 10 | 20 units | | Rs. 12 | 24 units |

Solution:

$$\Delta P = 12 - 10 = 2$$
$$\Delta Q_s = 24 - 20 = 4$$

$$PES = \frac{\Delta Q_s / Q_s}{\Delta P / P} = \frac{4/20}{2/10}$$

$$PES = \frac{0.20}{0.20} = \boxed{1.0}$$

Interpretation: - PES = 1.0 → **Unit Elastic Supply** - 1% price increase → 1% quantity supplied increase

---

Example 8B: Supply Elasticity (Percentage Method)

Problem Statement: *"Price rises by 8%. Quantity supplied rises by 16%. Calculate price elasticity of supply."*

Given: - % ΔP = 8% - % ΔQs = 16%

Solution:

$$PES = \frac{16\%}{8\%} = \boxed{2.0}$$

Interpretation: - PES = 2.0 > 1 → **Elastic Supply** - Supply is responsive to price changes - Producers can easily increase production

---

8. DETERMINANTS OF ELASTICITY

Determinants of PED:

Factor	More Elastic If...	Less Elastic If...
Substitutes	Many available	Few/None available
Necessity vs Luxury	Luxury	Necessity
Market Definition	Narrowly defined	Broadly defined
Time Horizon	Long run	Short run
% of Income	Large % of budget	Small % of budget

Determinants of PES:

Factor	Elastic Supply (PES > 1)	Inelastic Supply (PES < 1)
Capacity	Spare capacity available	Operating at full capacity
Stocks	High stock levels	Low/No stocks
Time Period	Long run (can build factories)	Short run (fixed capital)
Factor Mobility	Easy to employ factors	Requires specialized skills
Production Period	Quick to produce	Long production cycle

Example: Agricultural products - **Short-run:** Inelastic (takes 6 months to grow crops) - **Long-run:** More elastic (can plant more acres)

---

9. SUMMARY: ALL ELASTICITY FORMULAS

Elasticity	Formula	Interpretation	Example Use
PED	|ΔQd/Qd| / |ΔP/P|	>1: Elastic<1: Inelastic	Pricing strategy
YED	(ΔQd/Qd) / (ΔY/Y)	>1: Luxury<0: Inferior	Product classification
XED	(ΔQd,A/Qd,A) / (ΔPB/PB)	>0: Substitutes<0: Complements	Competitive analysis
PES	(ΔQs/Qs) / (ΔP/P)	>1: Elastic<1: Inelastic	Supply responsiveness

---

10. QUICK DECISION GUIDE

Question: "Should we raise price?"

Step 1: Calculate PED

Step 2: Apply TR rule: - If PED < 1 (Inelastic) → YES, raise price (TR ↑) - If PED > 1 (Elastic) → NO, lower price (TR ↑) - If PED = 1 → Price change won't help (TR same)

Question: "Are these goods substitutes or complements?"

Step 1: Calculate XED

Step 2: Check sign: - XED $> 0 \rightarrow$ **Substitutes** - XED $< 0 \rightarrow$ **Complements**

Question: "Is this a luxury or necessity?"

Step 1: Calculate YED

Step 2: Check value: - YED $> 1 \rightarrow$ **Luxury** - $0 <$ YED $< 1 \rightarrow$ **Necessity** - YED $< 0 \rightarrow$ **Inferior Good**
